USB Security Gateway

ESP32-P4NRW32-Based Hardware USB Threat Detection & Access-Control Gateway

Dual-USB proxy firewall with descriptor-based whitelisting and hardware power/data isolation on a 4-layer PCB

PCB Hackathon 2026 Submission

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1. IoT / Embedded Application Description

USB is the most common way for untrusted hardware to touch a computer. A keyboard-emulating device, a flash drive that also registers as a keyboard, or a device that draws abnormal power can compromise a host within seconds, because operating systems trust a device the moment it enumerates. Labs, kiosks, industrial control PCs and shared workstations need a small unit that sits in front of the host, decides what a device is allowed to be, and can physically cut it off.

2. Problem Statement

There is a need for a standalone, low-cost, hardware-enforced USB gateway that identifies a connected device from its descriptors before the host ever sees it, applies a whitelist and class policy, monitors the device's power draw, logs events independently of any PC, and can isolate the device in hardware (power and data) even if the gateway firmware fails - all on a manufacturable 4-layer PCB.

3. Proposed Solution

- An ESP32-P4NRW32 (U3) forms the central controller, chosen because it has a High-Speed USB 2.0 OTG host PHY and a separate Full-Speed USB OTG device PHY that can be used at the same time (plus a third USB-Serial-JTAG port for debug). This lets the unknown device and the PC sit on two independent USB links. It also provides hardware secure boot, flash encryption and cryptographic engines, and 32 MB of in-package PSRAM.
- Proxy-firewall architecture: the P4 acts as USB host to the untrusted device and as USB device to the PC. The PC never talks to the original device; it sees only a re-created, sanitized device that firmware chooses to present.
- Descriptor-based whitelisting: firmware reads the connected device's VID, PID, class, serial number and descriptors, compares them with a whitelist held in flash, and decides ALLOW, BLOCK, MONITOR or QUARANTINE.
- Hardware isolation: a TPS2553 load switch (U6) controls VBUS to the untrusted port (with current limit and fault flag), a TS3USB221A high-speed switch (U8) opens or connects the D+/D- lines, and two push-buttons (S2 and S3) force both off in hardware while held, even if the firmware hangs. Both switches default to OFF after reset.
- An INA219 current monitor (U7) on the untrusted port's VBUS flags abnormal power draw.
- An ATECC608B secure element (U14) verifies signed whitelist updates; the P4 itself adds secure boot v2, flash encryption, HMAC, ECDSA and a true random number generator.
- A 2.0-inch ST7789 TFT, four status LEDs, a buzzer and approve/block buttons give local feedback; a microSD card keeps an offline event log.
- A 5V USB input (J3) feeds the 5V_SYS rail through a Schottky diode (D1). An ISL80102 linear regulator (U1) makes 3.3V and a TPS7A1111 linear regulator (U2) makes the 1.1V core rail from 3.3V.

3A. What Makes This Design Different

Three design decisions set this board apart from a generic ESP32 development board:

1. Dual-USB Proxy Instead of a Pass-Through

The P4 cannot forward raw USB traffic through GPIO pins, and a true transparent High-Speed pass-through would need extra USB hub or bridge hardware. Instead the P4 terminates both sides: it is the host for the untrusted device (High-Speed PHY) and the device for the PC (Full-Speed PHY). This also means the PC only ever sees what the firmware lets it see, which is the security feature.

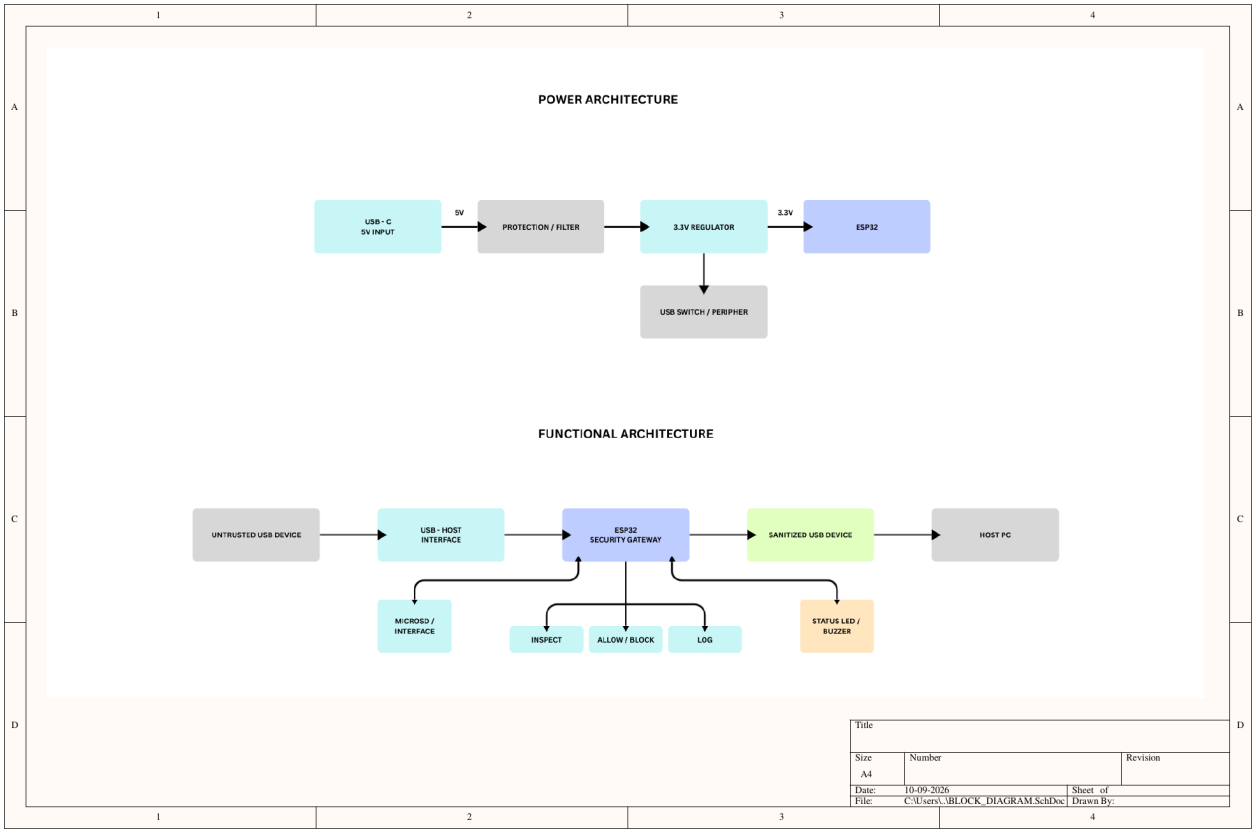
2. Layered Hardware Isolation

Blocking has four layers: firmware refuses to create the PC-side device, the U6 load switch cuts VBUS_A, the U8 data switch opens D+/D-, and the S2/S3 override buttons force both off in hardware. Cutting power alone does not stop a self-powered device, and cutting data alone leaves it powered, so both switches are used. The override is momentary: S2 pulls U6 EN to ground and S3 pulls U8 /OE high only while pressed. Note this is switching isolation, not galvanic isolation: grounds are shared.

3. Fail-Safe Defaults

U6's enable pin has a 10k pull-down (R22) and U8's output-enable pin has a 10k pull-up (R24), so after reset the port is unpowered and its data lines are open. The device receives nothing until firmware has made a decision and explicitly enables the port.

4. System Block Diagram



Block Diagram Explanation

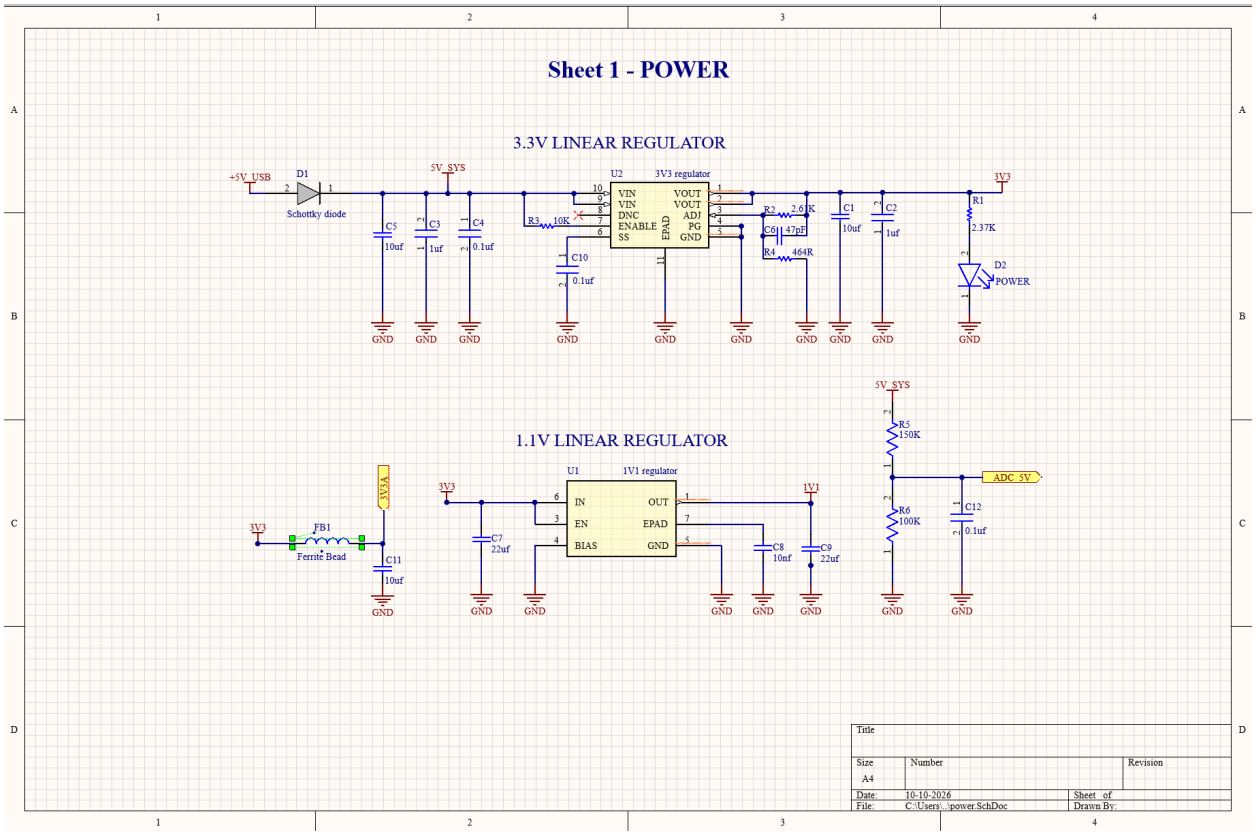
The diagram has two parts: the Power Architecture (top) and the Functional Architecture (bottom).

- **5V USB Input -> D1 -** 5V from the USB input (J3, net +5V_USB) passes through D1 (Schottky) to the 5V_SYS rail. R5/R6 divide 5V_SYS down to the ADC_5V sense net.
- **Linear Regulators (U1, U2) -** U1 (ISL80102) makes the 3.3V rail from 5V_SYS; U2 (TPS7A1111) makes the 1.1V rail for the P4 core from 3.3V. A ferrite filter (FB1) makes a quieter 3V3A rail for analog and USB PHY supplies. The 3.3V rail also feeds the USB switch and peripherals.
- **ESP32-P4NRW32 (U3) -** central processor; runs the USB host stack, the whitelist and policy engine (inspect, allow/block, log), and drives all isolation hardware, the display and logging.
- **J1 USB-A + U5 ESD -** untrusted-device port, protected by the low-capacitance (<=1 pF) ESD array U5 and a TVS (D5). R17/R18 15k pull-downs terminate D+/D-.
- **U6 TPS2553 Load Switch -** switches VBUS_A to the untrusted device, limits current and reports faults to the P4 (GPIO5 enable, GPIO6 fault).
- R23 + U7 INA219 - measures the untrusted port's current over I2C (GPIO7/8) so abnormal power draw can be detected.
- **U8 TS3USB221A Data Switch -** connects or disconnects the untrusted device's D+/D- from the P4 High-Speed PHY (GPIO9 controls /OE).
- **S2 / S3 Override Buttons -** hardware override: S2 forces U6 off and S3 forces U8 open while pressed, regardless of firmware.
- **J2 USB-C (to PC) -** Full-Speed device port (GPIO26/27). U9 protects the data lines; VBUS from the PC is used only to sense that a PC is connected (divider to GPIO17).
- **J3 (Power + Debug) -** J3 powers the board and exposes the P4's USB-Serial-JTAG port (GPIO24/25) for flashing, console and debugging.
- **ATECC608B, microSD, W25Q128JV Flash -** secure element for signed whitelist updates (U14), event log storage (J9), and firmware/whitelist storage (U4).
- **TFT, LEDs, Buzzer, Buttons, Tamper -** local status display, alerts, approve/block input and enclosure-tamper detection.

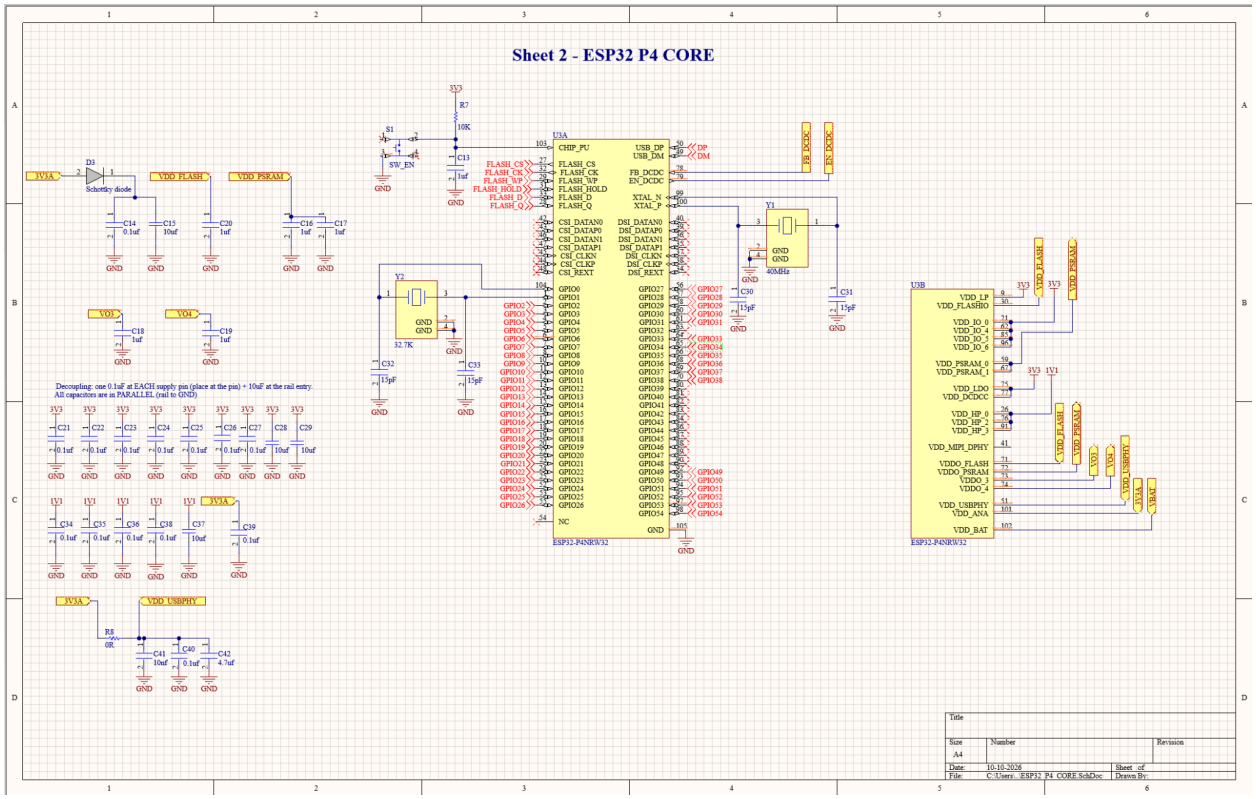
5. Schematic Design

The full schematic is split into 13 sheets following the hierarchy below. Power symbols are global nets; arrow labels connect to other sheets by name.

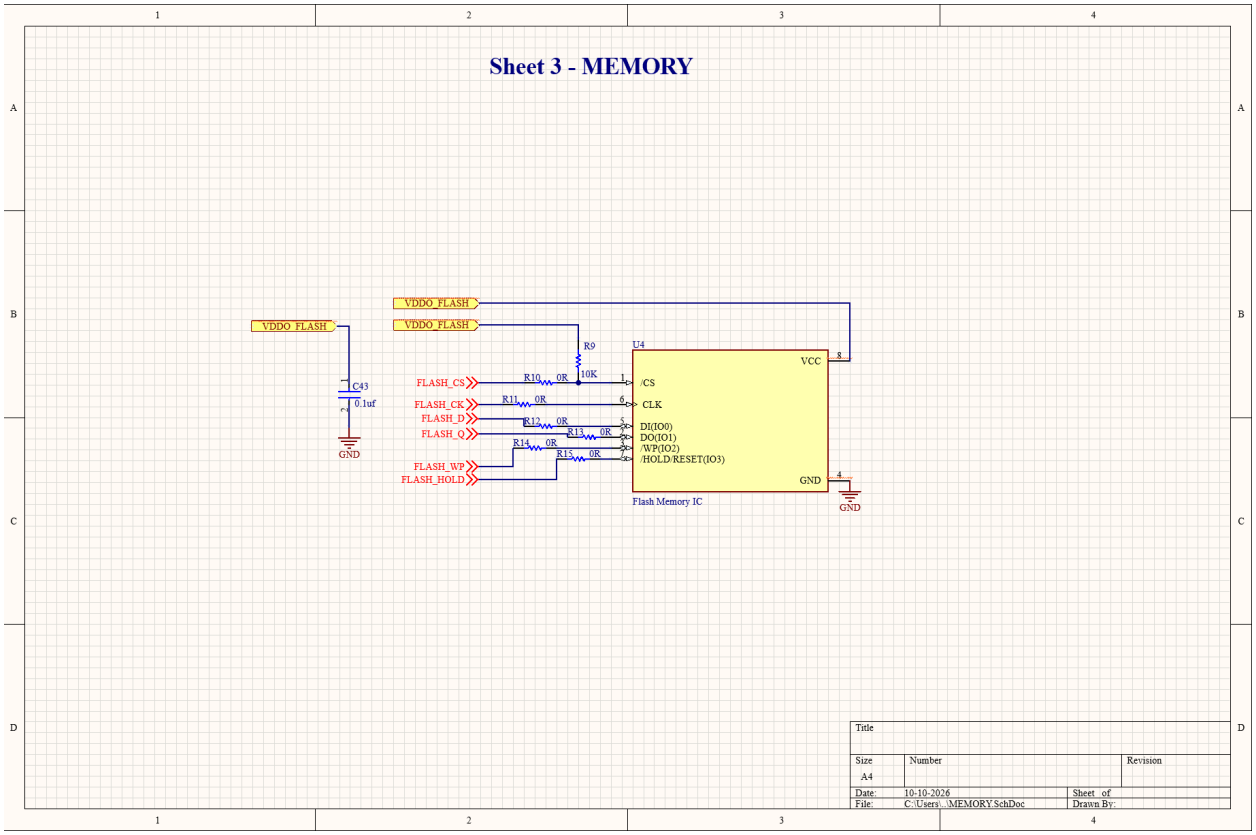
Sheet 1 - Power: 5V input diode, 3V3 LDO, 1V1 LDO, 3V3A filter, 5V sense



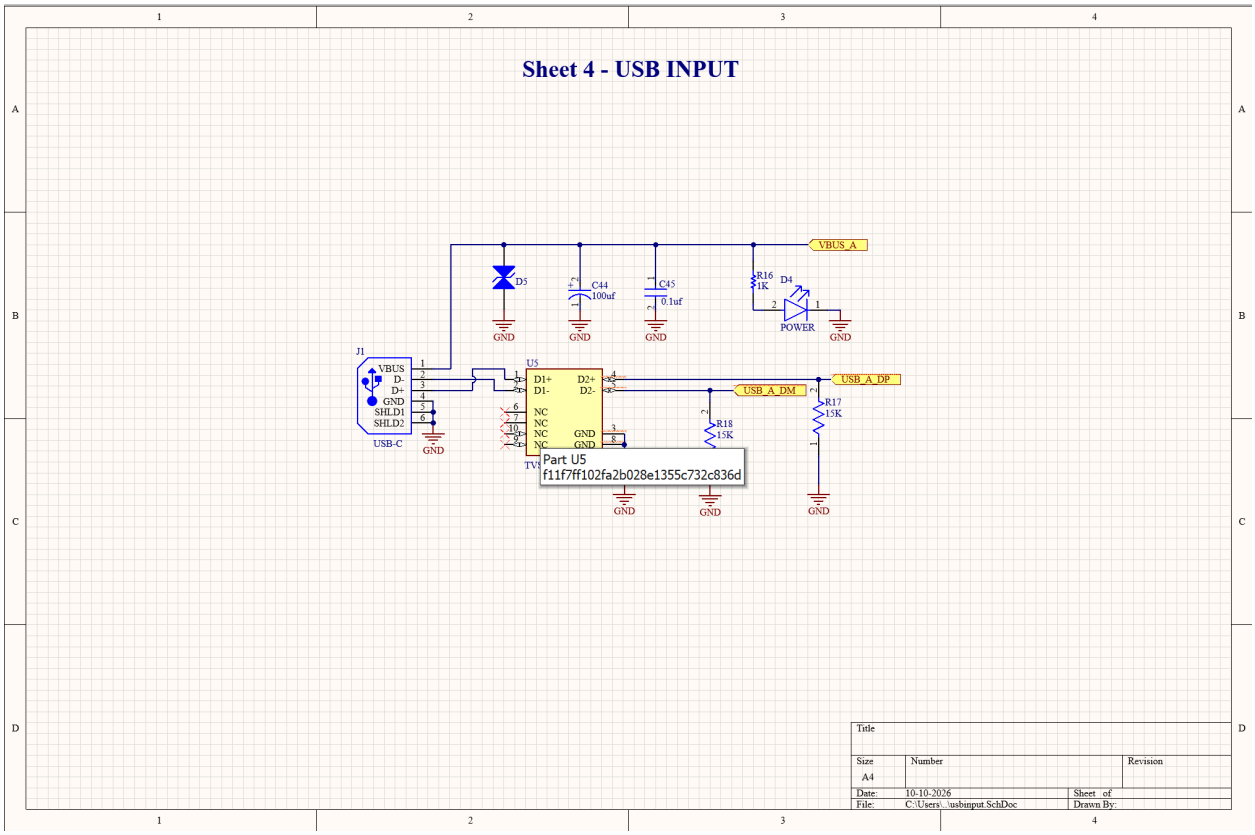
Sheet 2 - ESP32-P4 Core: power pins, per-pin decoupling, reset, 40 MHz and 32.768 kHz crystals



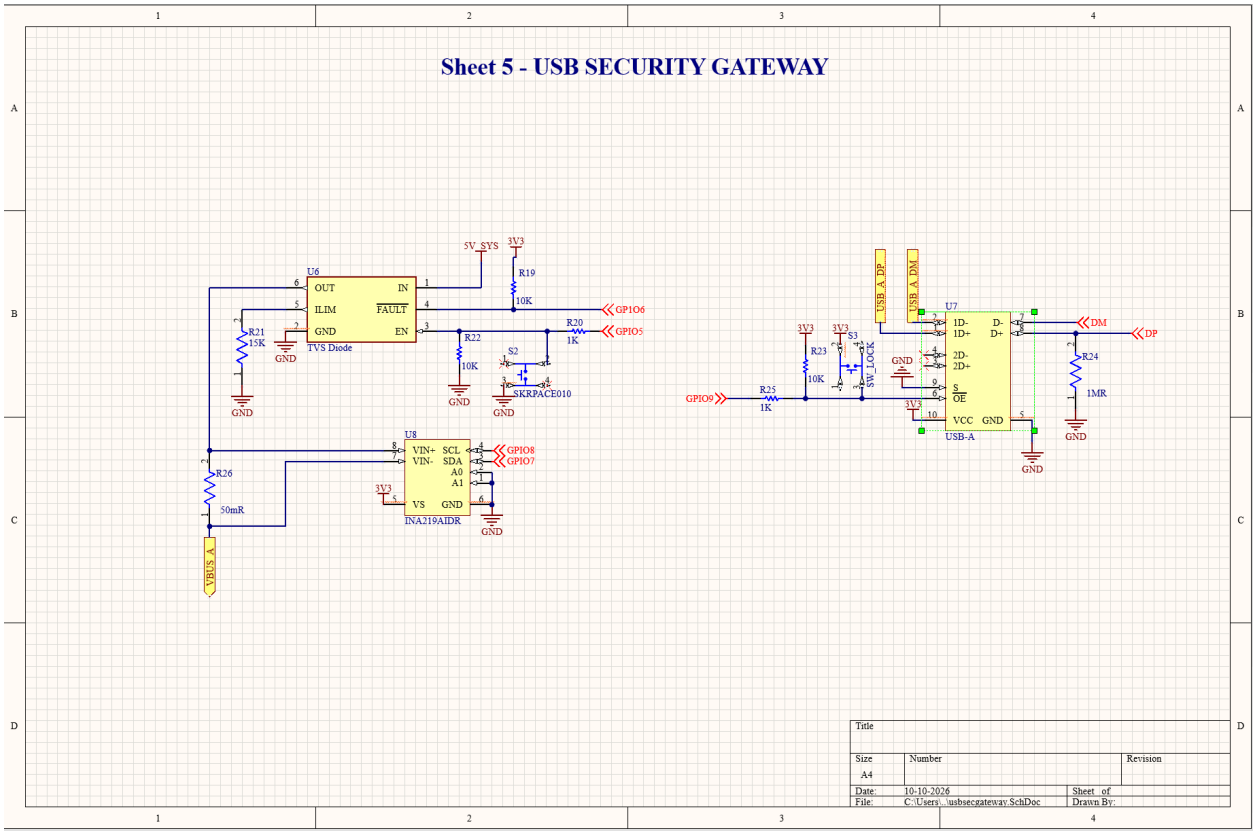
Sheet 3 - Memory: external QSPI flash



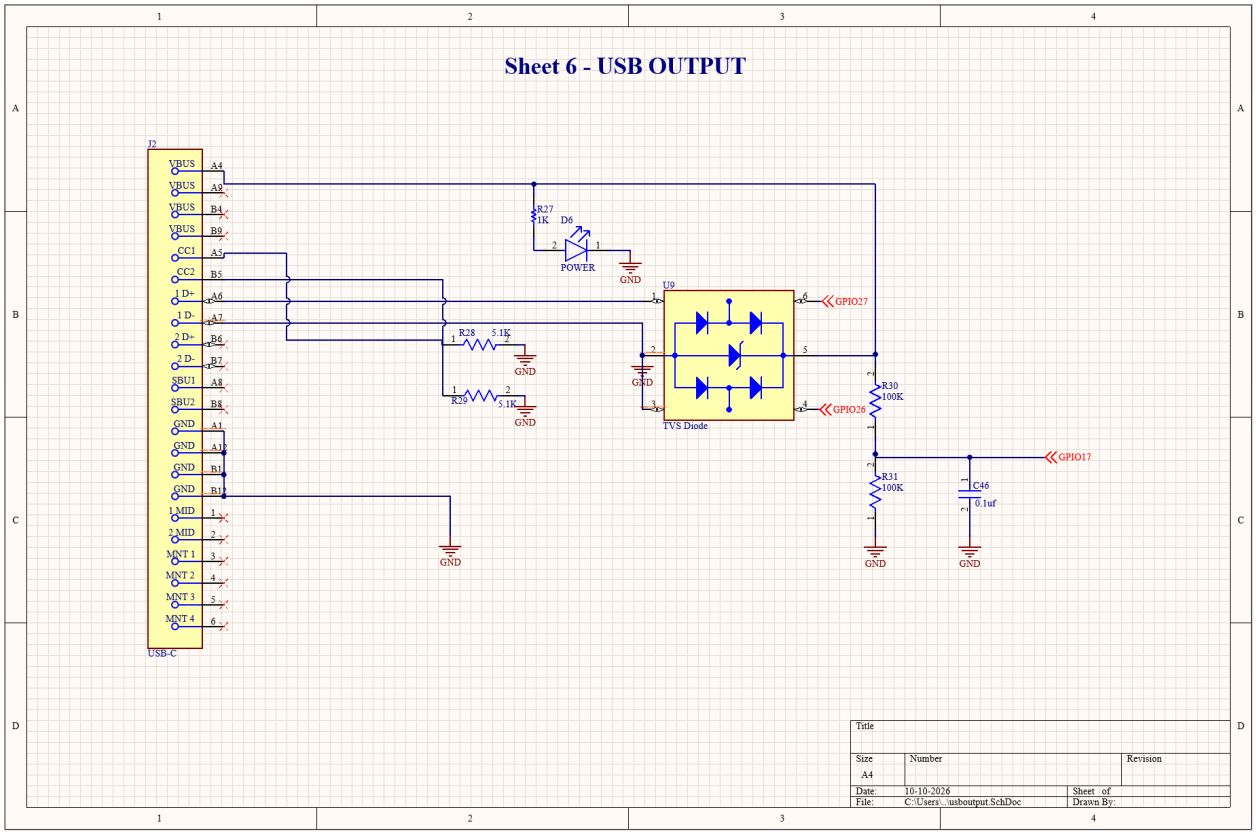
Sheet 4 - USB Input: untrusted USB-A port, Protection Diode, pull-downs



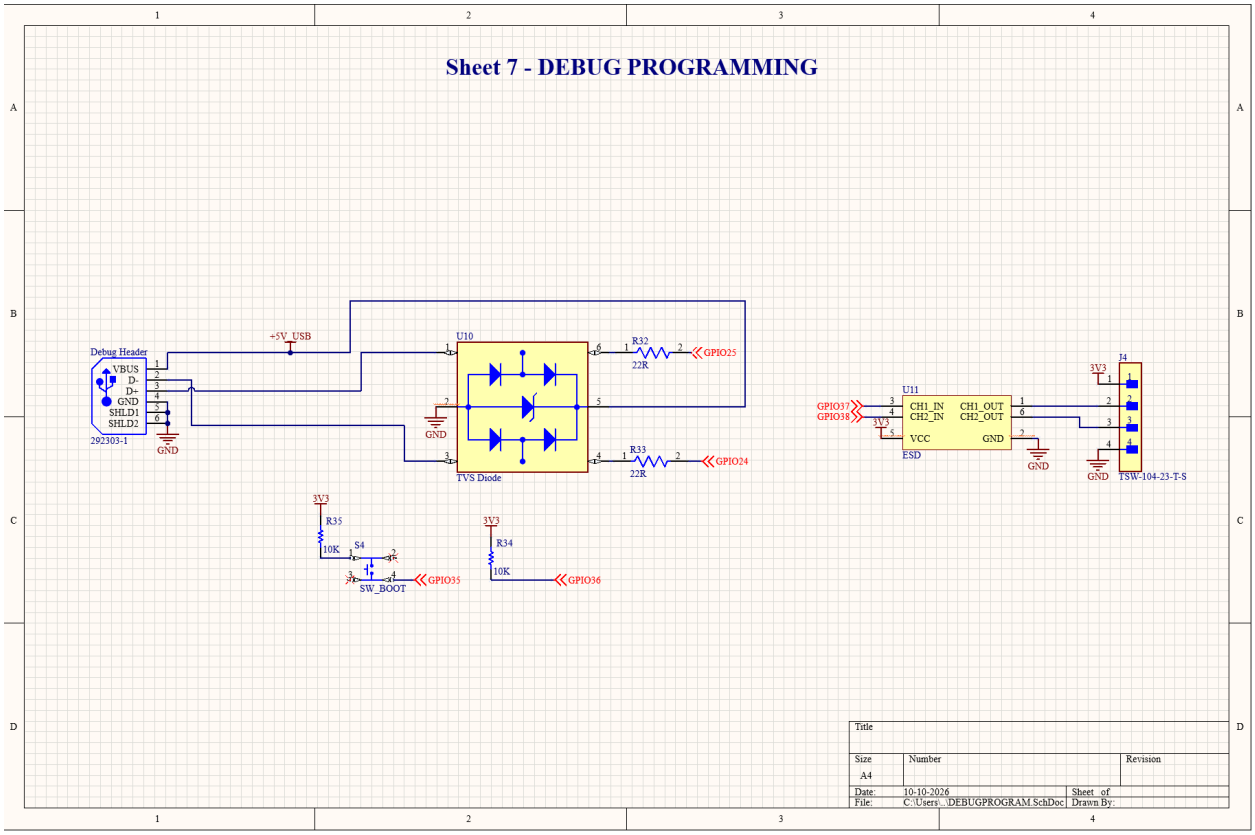
Sheet 5 - USB Security Gateway: VBUS load switch, current monitor, HS data switch, override buttons



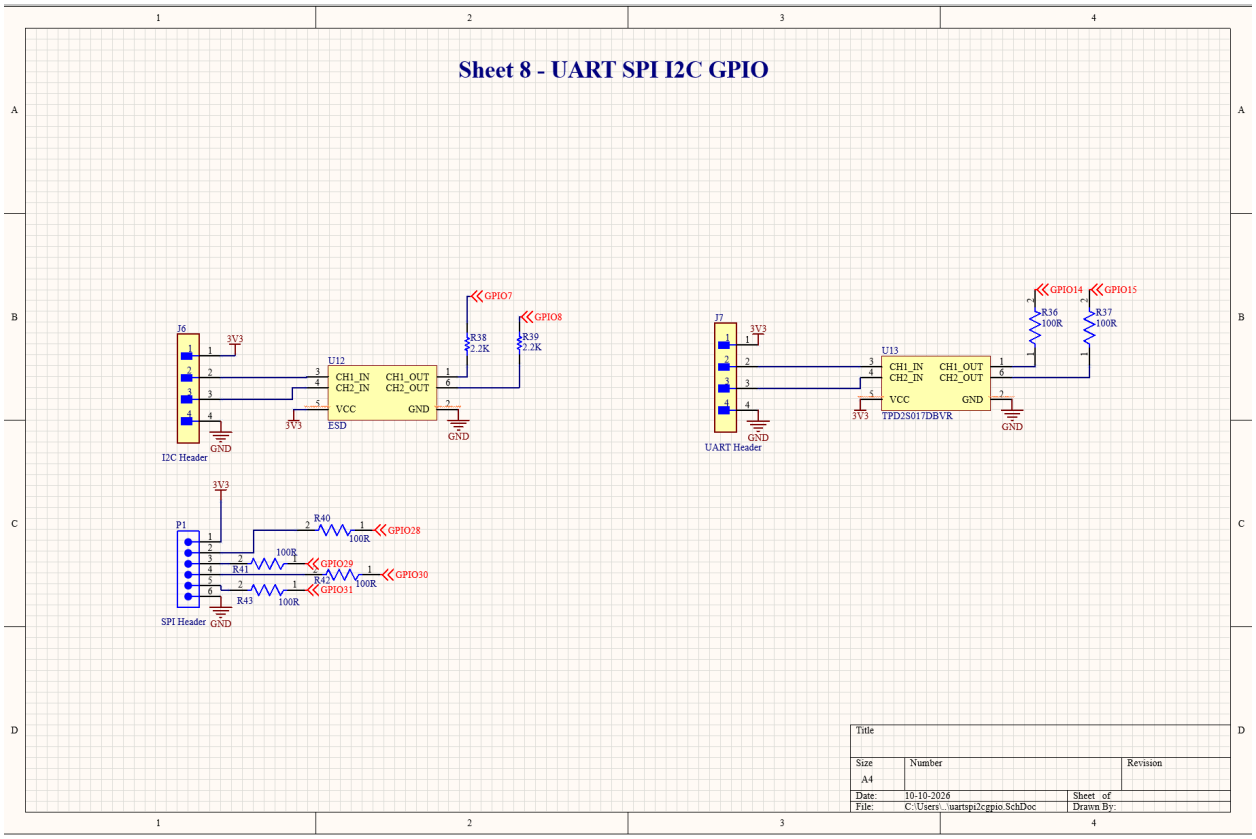
Sheet 6 - USB Output: USB-C port to the PC



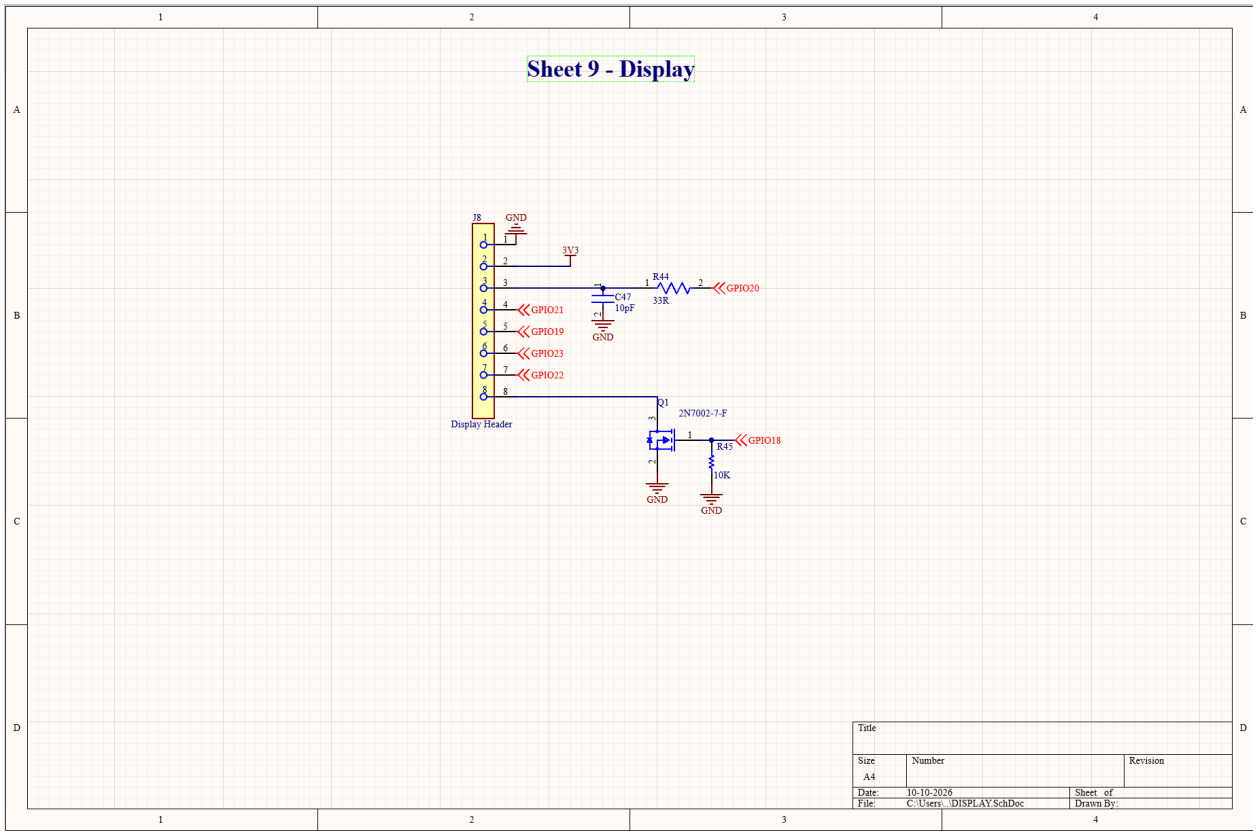
Sheet 7 - Debug & Programming: USB-Serial-JTAG port, BOOT, UART0 header, GPIO34 pad



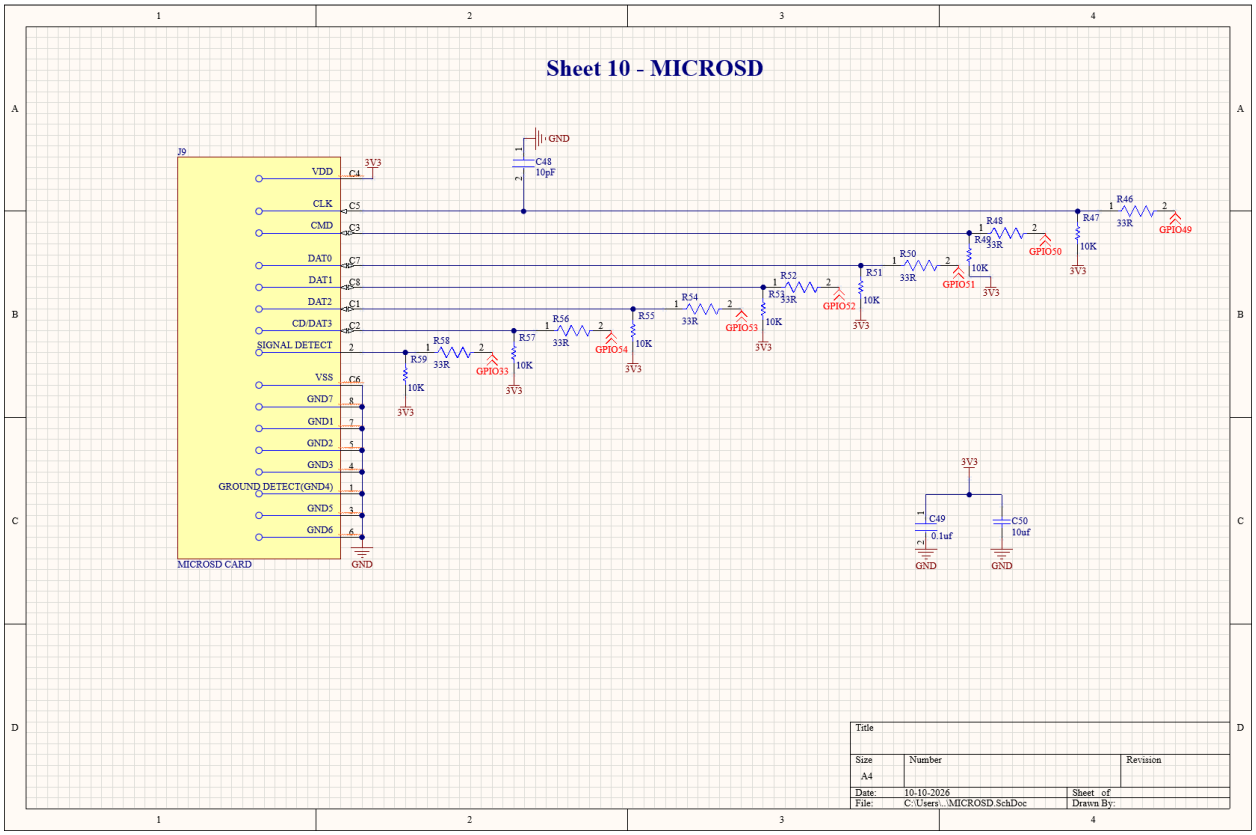
Sheet 8 - Communication: I2C, UART and SPI headers



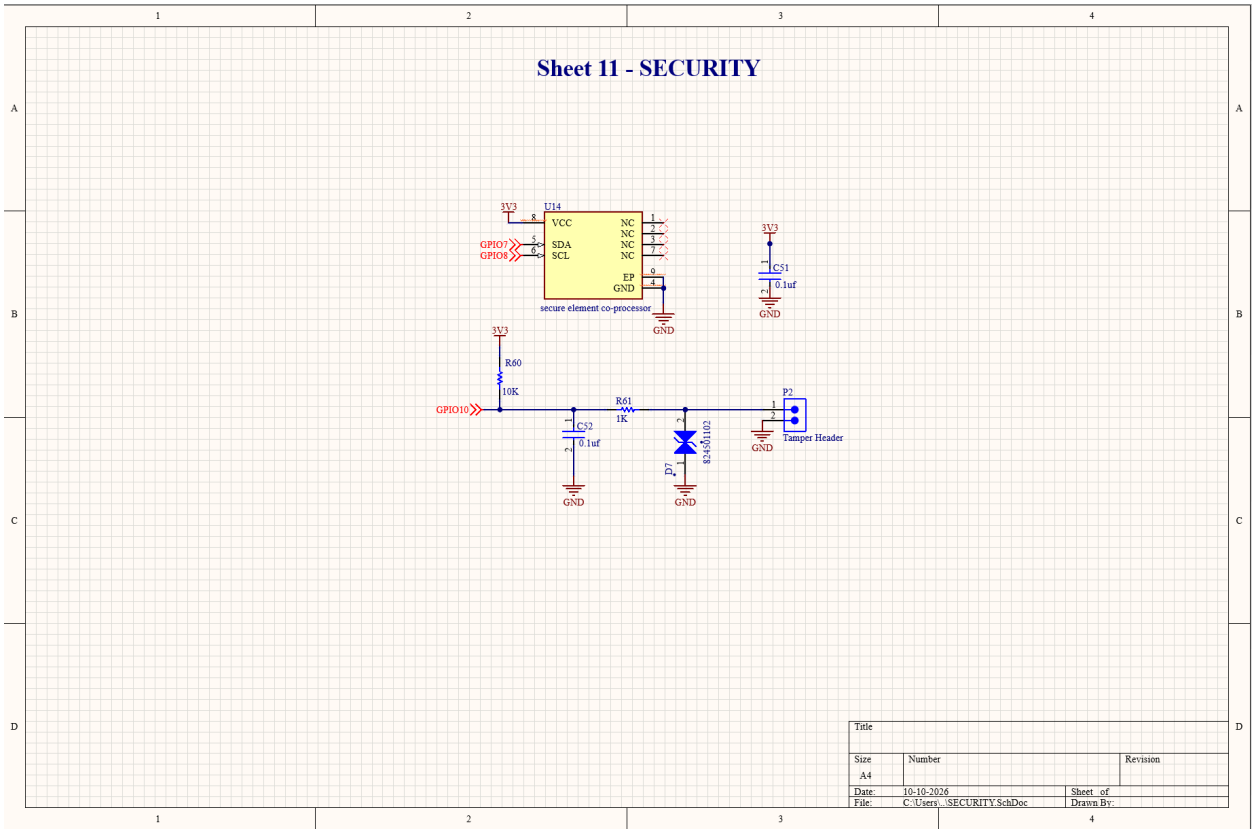
Sheet 9 - Display: ST7789 TFT



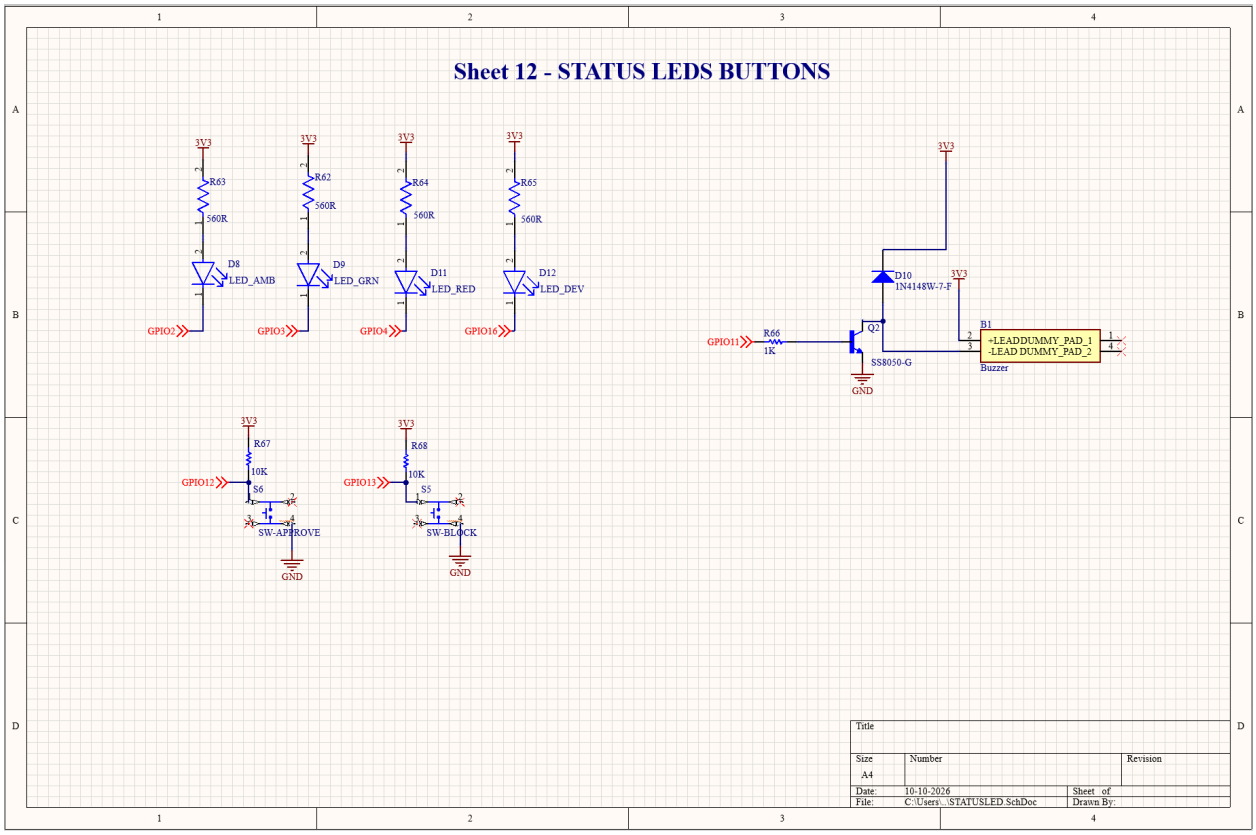
Sheet 10 - microSD: event logging



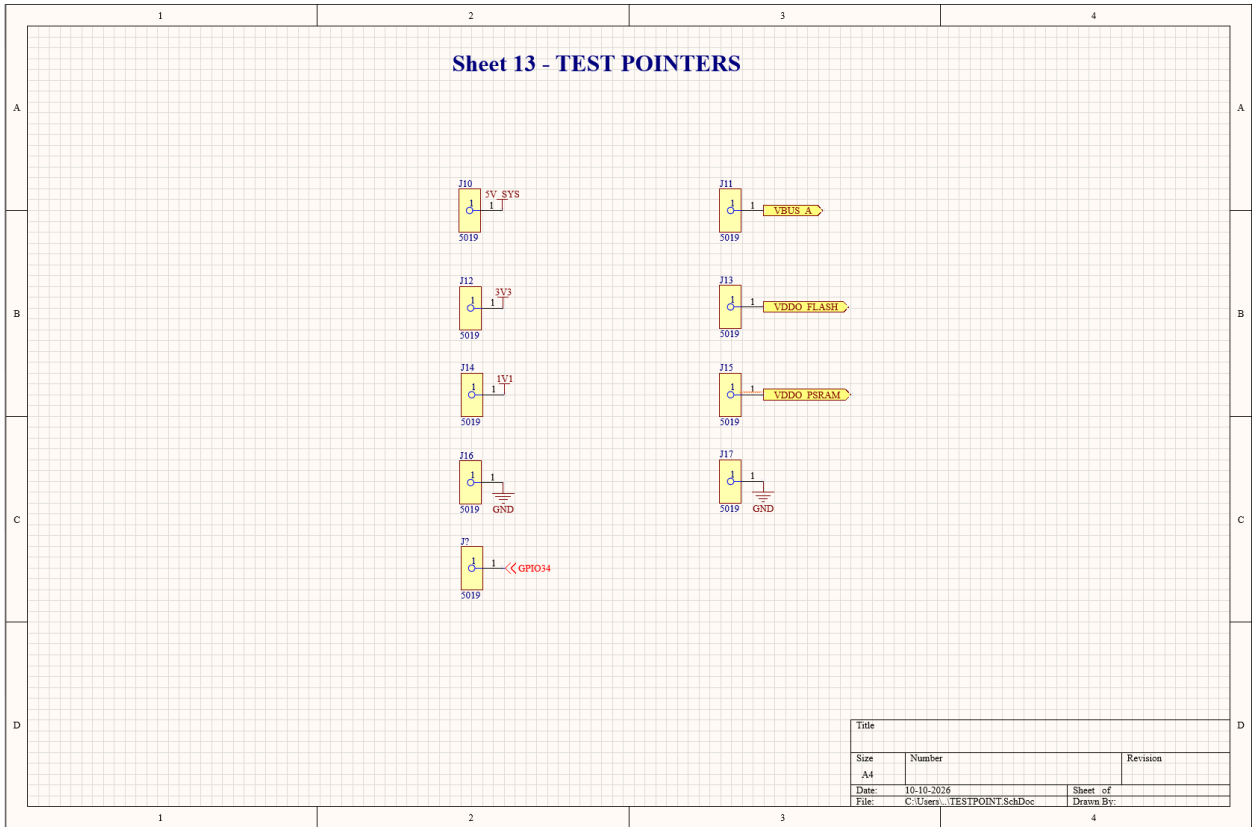
Sheet 11 - Security: ATECC608B, tamper input



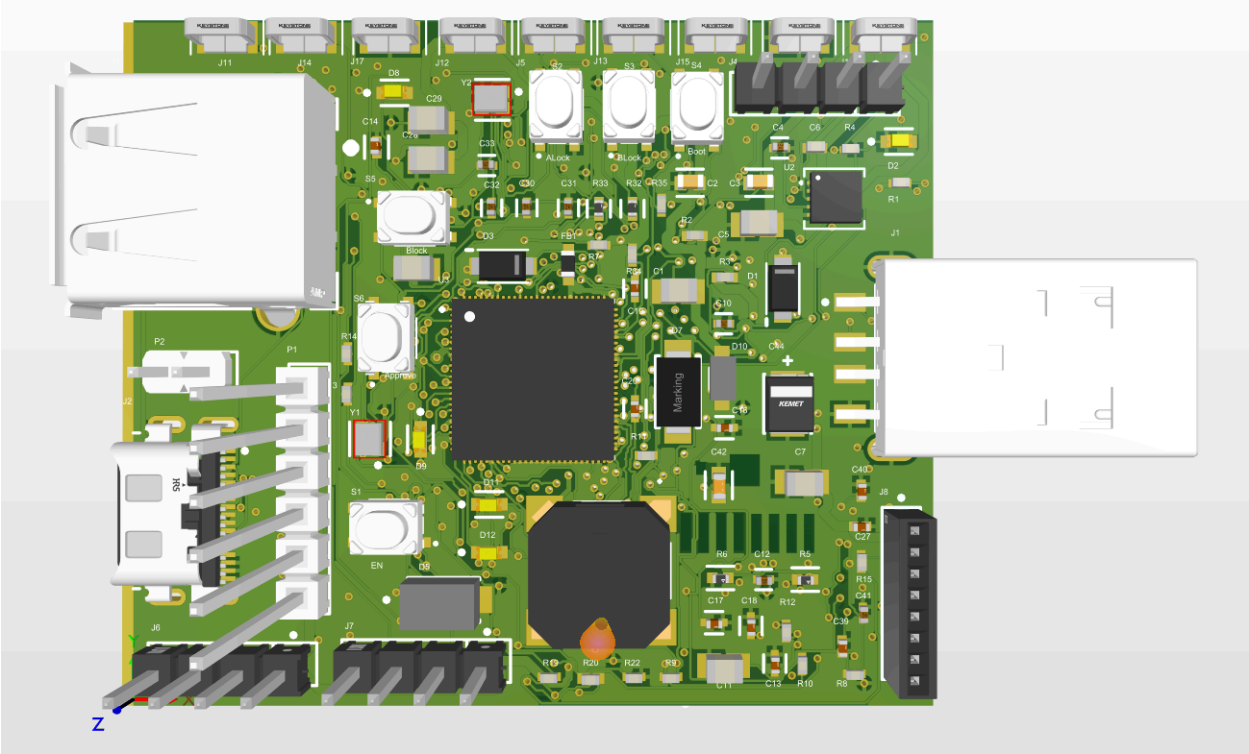
Sheet 12 - Status LEDs, Buttons & Buzzer



Sheet 13 - Test Points



6. 3D View



7. Module-Wise Component Notes (Design Justification)

This section captures, block by block, why each component was chosen - useful as a quick reference when explaining the design.

Block	Key Parts	Why This Part / Design Choice
ESP32-P4 Core	ESP32-P4NRW32 (U3A/U3B), Y1 40MHz, Y2 32.768kHz, S1 reset, R7 10k, C13 1uF	The P4 has a High-Speed USB OTG host PHY and a separate Full-Speed OTG device PHY usable at the same time, which makes the proxy-firewall architecture possible on one chip. It also has secure boot, flash encryption and crypto engines, and 32 MB in-package PSRAM, so no external memory bus is routed. Y1 is the main 40 MHz clock that also serves the USB PHY; Y2 (GPIO0/1) is optional for RTC timestamps. The reset button S1 pulls CHIP_PU low, with R7 pull-up and C13 for power-up delay. QFN-104 is the only package offered and its exposed pad gives low-inductance ground and heat removal.
External Flash	W25Q128JVSIQ (U4), R10-R15 0R, R9 10k pull-up, C43	The NRW32 variant has no in-package flash, so external QSPI flash is mandatory. 16 MB holds the firmware, two OTA images and the whitelist with room to spare. SOIC-8 is easy to hand-solder and probe. The 0 Ohm series links allow signal tuning; the CS pull-up (to VDDO_FLASH) keeps the flash deselected during reset.
5V Input & Protection	J3 USB receptacle (292303-1), D1 MBR140SFT1G, C3/C4/C5	J3 brings in 5V as +5V_USB. The series Schottky D1 stops current flowing back into the USB source and gives a small drop to 5V_SYS. The input has no PTC fuse or TVS, so it relies on the source and on D1 only. There are no CC resistors because J3 is not a USB-C connector. The source must supply the board load plus the untrusted port (up to about 0.5 A), and because the regulators are linear the input current equals the load current.

Block	Key Parts	Why This Part / Design Choice
3.3V LDO	U1 ISL80102IRAJZ, R2 2.61k / R4 464R, R3 10k, C6 47pF, C7 0.1uF, C1-C5, R1 2.37k + D2	A linear regulator removes the inductor and switching noise and keeps the layout simple. R2/R4 set the output; with a 0.5 V feedback reference they give about 3.3 V (verify against the datasheet). R3 pulls ENABLE up to 5V_SYS, C7 sets soft-start, and C6 is the feed-forward/compensation capacitor. The trade-off is heat: dissipation is (about 4.7 V - 3.3 V) x load current, so a 1 A load is roughly 1.4 W.
1.1V LDO	U2 TPS7A1111PDRVT, C10/C12 22uF, C11 10nF	The P4 core (VDD_HP) needs 1.1V. U2 is a 1.1V fixed-output LDO fed from 3V3 with EN tied high. Its 2.2 V drop turns every 100 mA of core current into 0.22 W in a small 2x2 mm package, so the real core current at 400 MHz should be measured. The output feeds the 1V1 net and test point J15.
3V3A Filter & 5V Sense	FB1 600R, C8 10uF, R5 150k / R6 100k, C9 0.1uF	FB1 and C8 make a quieter 3V3A rail for analog and USB PHY supplies. R5/R6 divide 5V_SYS by 0.4 (about 2.0 V at 5 V) to the ADC_5V net so firmware can watch the supply.
Untrusted USB Port	J1 USB-A (480-37-1000), U5 TPD4E05U06QDQARQ1, D5 TVS, C44 100uF, C45 0.1uF, R16 1k + D4, R17/R18 15k	USB-A matches the plugs on most devices. A High-Speed port needs an ESD array of 1 pF or less; common USBLC6 parts have too much capacitance for 480 Mbps. A bulk capacitor and TVS on VBUS_A absorb plug-in transients. The LED on VBUS_A is hardwired so it lights whenever the port is powered, independent of firmware. R17/R18 are the 15k host pull-downs on D+/D-.
VBUS Load Switch	U6 TPS2553QDBVRQ1, R19 10k, R20 1k, R21 15k, R22 10k, S2	Cuts power to the untrusted device, limits current (set by R21 on ILIM) and reports faults on GPIO6 (R19 pull-up). EN is driven from GPIO5 through R20 and has a 10k pull-down so power is off at reset. S2 shorts EN to ground to cut power by hand. SOT-23-6 is small and the load needs no heatsink.
Current Monitor	R23 50 mOhm, U7 INA219AIDR	A 50 mOhm shunt drops about 25 mV at 0.5 A and measures current over I2C (GPIO7/8, address 0x40 with A0/A1 grounded) with fine resolution. Used to detect abnormal power draw from the device.
HS Data Switch	U8 TS3USB221ARSER, R24 10k, R25 1M, R26 1k, S3	Opens the D+/D- lines between the device and the P4 for true data isolation, which VBUS switching alone cannot give for self-powered devices. The S pin is tied to ground so channel 1 is used and channel 2 is unused. /OE has a pull-up (R24) so lines are open at reset, and GPIO9 drives it through R26. S3 forces /OE high by hand. It sits in the 480 Mbps path, so it is the highest signal-integrity risk and must be kept short.
Hardware Override	S2, S3 (SKRPACE010)	Two momentary tact switches form the hardware lock. S2 forces U6 EN to ground and S3 forces U8 /OE to 3V3, both independent of firmware. Because they are momentary, the override holds only while the buttons are pressed.
PC Port	J2 CX90M-16P, U9 USBLC6-2SC6, R28/R29 5.1k, R30/R31 100k, C46, R27 1k + D6	Full-Speed device port to the PC. U9 protects the Full-Speed lines and goes straight to GPIO26/27 with no series resistors. CC pull-downs identify the board as a device. The PC VBUS is only sensed through the R30/R31 divider (ADC GPIO17) and never joined to the board supply, preventing back-feed. D6 shows when the PC supplies VBUS.
Debug & Programming	J3, U10 USBLC6-2SC6, R32/R33 22R, S4, R34/R35 10k, J4, U11 TPD2S017DBVR, J5	The P4's built-in USB-Serial-JTAG (GPIO24/25) handles flashing, console and JTAG without a separate USB-UART chip. GPIO35 with a pull-up and the S4 BOOT button selects download mode; GPIO36 is pulled up as required. The UART0 header J4 (GPIO37/38) sits behind U11 for ESD protection. J5 is a test pad on GPIO34.

Block	Key Parts	Why This Part / Design Choice
Secure Element	ATECC608B-MAHCZ-T (U14), C51	Hardware key store that verifies signed whitelist updates so the whitelist cannot be changed by an attacker. It shares the I2C bus (GPIO7/8) with the INA219 and the I2C header. The P4 already has ECDSA and HMAC, so this adds tamper-resistant key storage rather than new capability.
Display	J8 LPPB081NFFN-RC, R44 33R, C47 10pF, Q1 2N7002-7-F, R45 10k	Shows device identity and the allow/block decision. SPI is simple and fast enough; R44 and C47 on SCLK (GPIO20) damp edges. Q1 switches the backlight from GPIO18.
Event Logging	J9 DM3NW-SF-PEJ(800), 33R series, 10k pull-ups, C48-C50	Keeps an offline log independent of any PC. Runs on SDMMC slot 1 (GPIO49-54, card detect GPIO33) through the GPIO matrix at 3.3V, which is ample for logs. Every line has a 33 Ohm series resistor and a 10k pull-up.
Status Indicators	D8, D9, D11, D12, B1 MLT-8530, Q2, D10, S5, S6	Green/amber/red LEDs show ALLOW/MONITOR/BLOCK and D12 shows device presence. LEDs are active-low (3V3 through 560 Ohm to the GPIO) so they stay off during reset. Q2 drives the buzzer from GPIO11 with D10 as flyback diode. S5/S6 (GPIO13/12) give a local manual decision.
Tamper & RTC	P2, R60/R61, C52, D7 (824501102), Y2, D3	An enclosure switch on P2 (GPIO10, 10k pull-up, ESD diode D7) flags opening of the unit. The 32.768 kHz crystal keeps timestamps; VBAT is fed from 3V3A through D3, and no coin cell is fitted on this schematic.
Expansion Headers	J6, J7, P1, U12/U13 TPD2S017DBVR, R36-R43	Keeps the board useful as a general ESP32-P4 development board. J6 is I2C (GPIO7/8, 2.2k pull-ups), J7 is a UART pair (GPIO14/15, 100R) and P1 is a 6-pin SPI/GPIO header (GPIO28-31, 100R). U12 and U13 add ESD protection.
Test Points	J10-J24 (5019)	Pads for 5V_SYS, 3V3, 1V1, VBUS_A, VDDO_FLASH, VDDO_PSRAM, GPIO7, GPIO8, GPIO34, GPIO36, U0TXD, U0RXD and three grounds for probing during bring-up.

8. Working Principle

- On power-up, the 3.3V and 1.1V linear regulators bring up the P4. U6 and U8 are in their default state: untrusted-port power off and data lines open.
- Firmware boots, checks its configuration and whitelist, and waits for a device on J1.
- When a device is detected, firmware enables U6 (limited power) and then U8 (data path to the P4 High-Speed PHY).
- The P4 acts as USB host and reads the device descriptors: VID, PID, class, serial number and others. It checks them against the whitelist and class policy.
- The INA219 (U7) reports current on the port. Abnormal draw, repeated connect/disconnect, an unexpected class, or an unknown ID raise an alert.
- If the device is allowed, firmware presents a sanitized USB device to the PC through the Full-Speed port J2. If not, it disables U8 and U6 and logs the event.
- The decision shows on the TFT and LEDs, sounds the buzzer for a block, and is written to the microSD log. Signed whitelist updates are checked by the secure element.
- At any time, holding S2 forces port power off and holding S3 forces the data lines open in hardware.

9. Protection Circuitry & EMC/EMI Considerations

- Schottky diode D1 blocks reverse current on the 5V input. There is no PTC fuse or TVS on the input, so overcurrent and surge protection depend on the USB source.
- TVS D5 and bulk capacitance C44 on VBUS_A absorb plug-in transients on the untrusted port; U6 limits current and reports faults.
- Low-capacitance (<=1 pF) ESD array U5 protects the High-Speed pair without degrading the 480 Mbps signal; USBLC6 arrays U9 and U10 protect the Full-Speed and debug ports; TPD2S017 parts (U11-U13) protect the UART and I2C header lines; D7 protects the tamper input.
- Fail-safe defaults: pull-down on U6 EN and pull-up on U8 /OE keep the port off and data open at reset.
- S2 and S3 are an independent hardware override of both isolation switches while held.
- No back-feed: the PC's VBUS is only sensed through a divider, never joined to the board supply.
- Linear regulators produce no switching noise, which helps EMI, but they turn the voltage drop into heat, so U1 and U2 need copper area and a thermal check.
- Decoupling: one 0.1 uF capacitor at each supply pin (placed at the pin) plus bulk capacitors at each rail entry; all capacitors are in parallel from rail to ground.
- Series 22 Ohm resistors on the debug USB lines (R32/R33), 33 Ohm resistors on the microSD lines and an R/C damper on the display clock reduce edge noise. The Full-Speed PC-port lines have no series resistors.
- Planned 4-layer stack-up: L1 signal and components, L2 solid ground, L3 power and signal, L4 signal. USB High-Speed pair routed as a 90 Ohm differential pair, short and over unbroken ground, with ESD diodes at the connector. Crystals are kept close to the chip. (Design intent: layout is in progress.)

10. Bill of Materials (BOM)

Ref	Part	Package	Function
U3	ESP32-P4NRW32	QFN-104 10x10 mm	Main controller (32 MB PSRAM in-package)
U4	W25Q128JVSIQ	SOIC-8 208 mil	16 MB QSPI flash
U1	ISL80102IRAJZ	10-pin DFN	3.3V adjustable LDO
U2	TPS7A1111PDRVT	WSON-6 2x2 mm	1.1V LDO for VDD_HP
U6	TPS2553QDBVRQ1	SOT-23-6	VBUS load switch with current limit
U7	INA219AIDR	SOIC-8	VBUS_A current monitor (I2C)
U8	TS3USB221ARSER	UQFN-10	High-Speed USB data switch
U14	ATECC608B-MAHCZ-T	UDFN-8	Secure element
U5	TPD4E05U06QDQARQ1	USON-10	Low-capacitance ESD array for HS pair
U9, U10	USBLC6-2SC6	SOT-23-6	ESD protection for PC port and debug port

Ref	Part	Package	Function
U11, U12, U13	TPD2S017DBVR	SOT-23-6	ESD protection for UART0, I2C and UART1 headers
D1, D3	MBR140SFT1G	SOD-123FL	5V input Schottky, VBAT OR diode
D5	TVS (value not shown on sheet)	To be confirmed	VBUS_A surge protection
D7	824501102	To be confirmed	ESD protection on tamper input
D10	1N4148W-7-F	SOD-123	Buzzer flyback diode
D2, D4, D6	Power LEDs	0603	Hardwired 3V3, VBUS_A and PC-VBUS indicators
D8, D9, D11, D12	Amber, green, red, device LEDs	0603	Status indicators (active-low)
Q1, Q2	2N7002-7-F, SS8050-G	SOT-23	Backlight driver, buzzer driver
FB1	600 Ohm ferrite	0603	3V3A filter
Y1	40 MHz crystal	3.2x2.5 mm	Main clock
Y2	32.768 kHz crystal	3.2x1.5 mm	RTC clock (optional)
J1	480-37-1000 USB-A	Through-hole	Untrusted device port
J2	CX90M-16P USB-C 16-pin	SMD + THT legs	PC port
J3	292303-1 USB receptacle	Through-hole	5V power and debug port
J9	DM3NW-SF-PEJ(800)	SMD	microSD socket
J8	LPPB081NFFN-RC	8-pin header	Display connector
J4, J6, J7	TSW-104-23-T-S	4-pin header	UART0, I2C and UART1 headers
P1	TSW-106-17-G-S-LL	6-pin header	SPI/GPIO header
P2	MTLW-102-05-G-S-172	2-pin header	Tamper switch input
B1	MLT-8530 buzzer	8.5x8.5 mm SMD	Audible alert
S1-S6	SKRPACE010 tact switch	SMD	Reset (S1), override (S2, S3), BOOT (S4), approve/block (S5, S6)
J5, J10-J24	5019 test point	SMD pad	Test points
R (all)	1 k, 10 k, 15 k, 22/33 Ohm, 5.1 k, 100 k, 150 k, 1 M, 2.2 k, 560 R, 0R, others	0402 (1%)	Pull-ups, series, CC, dividers, LED limiters
R23	50 mOhm 1%	0805	VBUS_A current shunt
C (0.1 uF, 1 uF, 10 nF, 15 pF, 10 pF, 47 pF)	X7R / C0G (crystal caps)	0402	Decoupling, crystal loads, filters
C (10 uF, 22 uF)	X5R 10V	0805	Bulk and LDO input/output
C44	100 uF polymer/tantalum	~6.3x5.4 mm	VBUS_A bulk

Note: packages are taken from the part numbers and typical choices and must be confirmed against each datasheet and the Altium BOM export.

Package Selection Rationale

- QFN-104 (MCU): the only package offered for the ESP32-P4; its exposed pad is the ground and heat path.
- SOT-23 / SOIC-8 / UQFN / WSON for ICs: small enough for a compact board and still stocked by assembly houses.
- 0402 for small passives next to the MCU: lowest inductance at the supply pins; 0603 for LEDs and ferrite for easy rework.
- 0805 for 10-22 uF capacitors: larger bodies lose less capacitance under DC bias.
- Through-hole for USB-A, the J3 power/debug connector and headers: these connectors take mechanical stress from repeated plugging.

11. Scope & Honest Limitations

- Detection is descriptor-level: the gateway identifies devices by VID/PID/class/serial and descriptors and watches power draw and connection behaviour. It does not scan file contents or detect malware inside a device.
- It does not remove malware; it isolates the device. A device that copies a whitelisted VID/PID is harder to spot, so the serial number and full descriptor set are checked too.
- Transparent High-Speed pass-through, USB packet capture and SuperSpeed are not supported by the ESP32-P4 alone and would need additional hardware.
- The PC-side link is Full-Speed (12 Mbps); the untrusted side can be High-Speed. Classes that cannot be proxied (for example video or audio) are blocked by default.
- The ESP-IDF host stack has no hub Transaction Translator support, so the untrusted port takes one directly attached device and no hub.
- Isolation is switching isolation, not galvanic: grounds are shared.
- The hardware override (S2/S3) is momentary. It does not latch the port off, so a permanent lock must come from firmware.
- The board has linear regulators only: efficiency is low and heat grows with load, so the supply must be checked at full load.
- There is no input fuse or surge clamp, and no network alerting: alerts go to the display, LEDs, buzzer and microSD log.
- Wi-Fi and Bluetooth are not on the P4; they would need an ESP32-C6 companion chip.